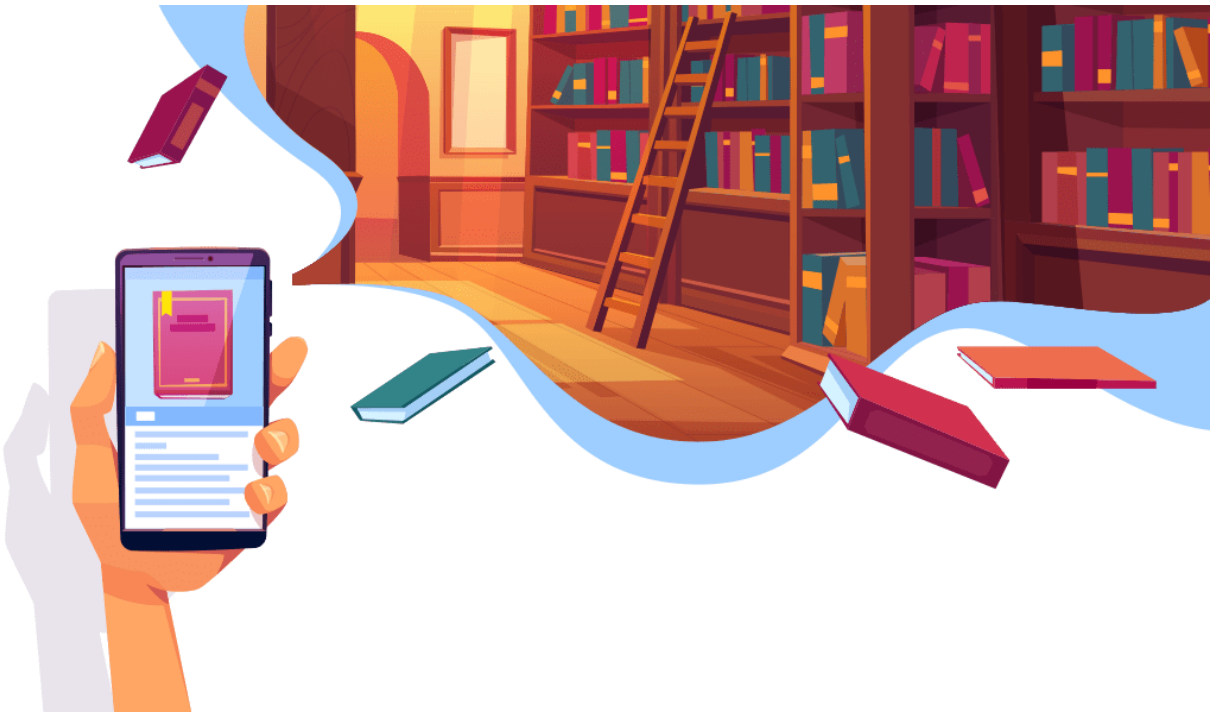
Course Project

User guide for Library Management System

COS3040A_2026SP: Programming in Python



Made by: Dzhano Mihaylov

Student ID: 200203987

Instructions:

1. Run the program as you would any other Python Console project

2. When the program starts, you will be greeted by a welcome message similar to those from old-school programs.

```
Welcome to our library program! Here you can upload the collection of items contained in your library.
In order to add items to the system you need to fill the "items.txt" file with the information regarding each physical and electronical item or
put your file following the following instructions.
WARNING! Read the instructions beforehand in the above mentioned "items.txt" file to avoid any unnecessary program exceptions. Also, please fill
 an appropriate data (no negative years, unallowed conditions, etc.).

Thank you for buying our program!
If there are any problems, feel free to contact our support team :)
```

   a. You will be prompted to read the instructions from the “items.txt” file to prepare beforehand the file with the items for the library in the appropriate format – whether it is “items.txt” or another.

```
items.txt
# This is the file where the items can uploaded according to their types.
# You must follow the standard protocol on how the information for each of them must be uploaded (order, number of data, type, etc.)
# If the item type is not one of the following - Book, Ebook, Magazine, DVD or Software - the program will not accept it.

# Date input format for:
# Book, "name", ID, "description", author_names, genre, year, pages, release_date, reservation_date, return_date, condition

# Ebook, "name", ID, "description", author_names, genre, year, size_in_MB, release_date, reservation_date, return_date, expiration_date

# Magazine, "name", ID, "description", publisher, issueNumber, release_date, reservation_date, return_date, condition

# DVD, "name", ID, "description", director_names, genre, year, duration, releaseDate, reservation_date, return_date, condition

# Software, "name", ID, "description", developer, username, subscription_plan, usage_time, release_date, reservation_date, return_date, expiration_date

#

# Warning: Names of the authors should be in the following format: firstName lastName   # Otherwise the program can crash.
#          Year of the physical item should not be negative number! It should be the same year as the one in release date.
#          Condition of a physical item should be from the following: New, Good, Average, Bad, OnTheVergeOfCollapsing or Stolen
#          Subscription of an electronical items should be from the following: Free, Basic, Standard or Premium
#          If a specific data is missing, then the program may crash!
# Please upload correct information!
```

      i. If any of the items is not in the correct format or its data is not accurate (future year, etc.), the command window will display a message for each problem for the respected item and will not create it to avoid conflicts.

```
There is already an item with this ID: 9780061120084. Therefore, the new one shall be skipped.
The release date of the item with ID 19 is not in the correct format! Therefore the item will not be created!
The year of the DVD with ID: 1 cannot be different from the release date's one. Therefore the item will not be created!
```

3. You will be asked if you would want to use “items.txt” or another file.

```
What file would you want to use? "items.txt" or another? Answer explicitly:
```

   a. If you choose “items.txt” the program continues to the next step (4.).

b. If you choose “another”, you will be asked for the name of the file in the correct format

```
What file would you want to use? "items.txt" or another? Answer explicitly: another
Please enter the location of the file with all of the necessary parts. Including slashes (/) if the file is not in the same folder as "items.txt".
As well as the ".txt" so that it can be read by the program.
File:
```

i. Before you confirm the name of the new file, you will be asked if you are sure you want to use that file.
   - If you choose “Yes”, you will continue to the next step (4.).
   - If you choose “No”, you will be asked again for the new file’s name:

```
Are you sure this is the file you want to use? Yes or No (answe
explicitly, if you do not, the program will ask you again): No
Enter the file name again:
```

4. Afterwards, you will be given two options about the way you want to upload commands.

```
How would you like to send commands to the program?
With the "commands.txt" file (beforehand prepared) or from the command window directly (at the moment)?
Answer explicitly: "File" or "Window":
```

a. If you choose “File”, all commands in “commands.txt” file will be executed one after another until the last line or the input “End program”.
   i. If you want to use this, beforehand you must have prepared the “commands.txt” file so that all commands are written correctly and run smoothly.

b. If you choose “Window”, you can pick commands on the go. Allowing you to completely interact with the items in real time.
   i. Before you start with writing commands, you will be show on the window the instructions on what you can write so that you can display and/or edit the items.

```
You can enter the following types of commands:

"CheckIfLate {MM/DD/YYYY}" - checks whether each item is late depending on the provided date
"DegradeCondition all/{specific ID} ./{type of Condition}" - degrades the condition of a specific physical item or all of them
"CalculateLateFee {specific ID}/all {MM/DD/YYYY}" - calculates the late fee of a specific item or all of them
"GetCodeExpirationDate {specific ID}/all" - gets the expiration date of the electronic item(s) code after which it won't work anymore
"GetsStolen {specific ID}/all" - a specific physical item or all of them are considered stolen from the library
"Display physicals / electronics / books / ebooks / magazines / DVDs / softwares / late / all / everything" - displays on the command window
the desired output of data
"Show physicals / electronics / books / ebooks / magazines / DVDs / softwares / late / all / everything" - displays on the command window the
 desired output of data

"Show commands" - display the current commands again. Useful if you do not want to scroll all the way up to see what functions you can do.
"End program" - stops the acceptance of new commands
Anything else written will not be accepted and the program will ask you for a new commands without doing anything beforehand.


Enter a command:
```

c. If you do not write any of those, you will be asked again until you pick one option.

```
Answer explicitly: "File" or "Window": neither
Please select "File" or "Window": no
Please select "File" or "Window": hey
Please select "File" or "Window": Hello Professor Mitreva ;)
Please select "File" or "Window":
```

5. After executing all commands, you will be asked if you would want to export the data for all items in “output_data.txt” file.

```
Would you want to export the items' data in another file? Yes or No
(answer explicitly - if not, it will be considered as "No"):
```

   a. If you choose “Yes”, the file will be rewritten with the data of the items that were added.
   b. If you choose “No” or write anything else, nothing will happen and the file will stay the same.

6. In the end, there is a farewell message by the creator thanking you for using the program.

```
Thank you for using my program! Have a great day!
Creator: Dzhano Mihaylov
Student ID: 200203987
Course: COS3040A_2026SP Programming in Python
Professor: Emanuela Mitreva
```